

Software Engineering

Introduction:

Defn

Software engineering is the branch of computer science, that deals with design, development, testing & maintain the software application, is called as the software engineering.

The software consist of different disciplines like ① Project planning ② Project management ③ Systematic analysis design ④ Validation & ⑤ Maintenance.

IMP

Goals of S.E.

- ① The primary goal of software engineering is to improve the quality of software.
- ② To increase the productivity of software.
- ③ Job satisfaction in software engineering.

IMP

Characteristics of S.E.

- ① The software is developed fundamentally but it is different from manufacturing process, in both process there are some of similarities & dissimilarities, high quality is achieved in both the process depends on customer requirement. The requirement are changes, processes also changes.
- ② Software is deterministic because it does not wear out. The software is deterministic but hardware is probabilistic.

Hardware has high value rates because of poor quality, design error & overload of components. The efficiency of hardware is decreases day by day but not software.

③ Software is customized

The software are developed according to customer requirement. So, software is customized.

④ Versions created.

The new version of software is formed by new modification & upgradation in existing software.

⑤ Hardware requirement

Different hardware component are used to designing of system & also enterchanging because of wear out.

IMP Qualities of Software

- | | |
|-----------------|--------------------------|
| ① Correctness | ⑥ Mantance |
| ② Reliability | ⑦ Re-usability |
| ③ User friendly | ⑧ Portability / Portable |
| ④ Robustness | ⑨ Productivity |
| ⑤ Performance | ⑩ Visibility |
| | ⑪ Interoperability |

There are many software qualities, some are related with product & some are related with processes use to create the software.

Qualities of S.E

① Correctness

If the software performs any task or operation according to the requirement then the output is functionally correct.

The correctness of software is accessed systematically. The correctness is also called as accuracy of the system.

② Reliability

Reliability is the probability that the software will operate correctly in a specified time interval.

The software can perform the given task in even time duration.

The measurement unit is MTBF (Mean time before failure)

③ User friendly

User friendly means the new user easily operate the system is called as user friendly.

④ Robustness

A software system always does not accept perfect input & output of hardware components. Robustness means the software continues operate correctly at the time of invalid input.

⑤ Performance (Efficiency)

The performance of software depends on the execution speed & memory requirement.

The performance of software increases

when maximum instruction are executed in minimum time duration & less memory req.

⑥ Maintenance -

Maintenance of software related with ablation addition of interchanging component in system as well as remove in the errors in the systems. There are two types of ^{software} system maintenance & hardware maintenance. software maintenance means we have providing updation or agration in software & it must be accepted by system & working continuously. hardware maintenance means we have interchange one hardware component to another component.

⑦ Re-usability

Reusability means the software modify by some small changes & it is use in new system then it is re-usable.

⑧ Portability | Portable

The software can be run in different environment then it is portable, the software easily transfer from one computer from another computer system. The portability means software can be run in different kind of computer system like specific RAM size, memory & operating system.

⑨ Productivity

Productivity is related with software production if software is developed correctly in less time, then the productivity is increases.

⑩

Visibility

The software development process should be transparent due to visibility all development team members & customer know the exact status of software development process & also customer requirement full fill or not.

⑪

Interoperability

Interoperability means the ability of system to for exist & cooperate with other system.

example,

- ① A computer system exactly ternally connected different sensor devices.
- ② In ms-office we are working in ms-word at that time we directly copy the data in ms-Excel that is called as interoperability, between the ms-word & ms-Excel.